

PRATEEK KOMARLA

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EDUCATION

Rutgers University

New Brunswick, NJ

Bachelor of Science in Computer Science and Data Science

May 2028

- Relevant Courses: Data Structures (Java), Linear Algebra, Discrete Structures, Computer Architecture, Data Management for Data Science, Design and Analysis of Computer Algorithms, Numerical Analysis and Computing
- GPA: 3.79 out of 4, Dean's List

PROFESSIONAL EXPERIENCE

Neuzen AI

May 2026 – August 2026

Software Engineer Intern

Dallas, Texas

- Designed a Model Context Protocol server for an AI-powered social media content generation platform.
- Gave AI assistants (Claude, ChatGPT, Cursor) tool-based access to generate, review, and publish content – 15 tools spanning content generation, live publishing, and multi-day campaign planning with a human-approval workflow.
- Built and designed a token and OAuth-based authentication architecture, to produce a full technical implementation for MCP connection.
- Built a REST API and three Agent Skills, giving autonomous coding agents direct platform access.

Rutgers University

Jan 2026 – Currently Working

Research Assistant / Computer Vision & Deep Learning for Biomedical Image Analysis

Piscataway, NJ

- Gathered numerical data from images regarding scratch area over time and conducted statistical analysis in Excel.
- Built a Python program to conduct image color-thresholding analysis to gather data for biological experiments.
- Built a computer vision and deep learning system, implementing a U-Net convolutional neural network from scratch using TensorFlow/Keras to achieve end-to-end semantic segmentation of wound regions across 1M+ pixels.

Astralinx

May 2025 – August 2025

Software Engineer Intern

Remote

- Developed a business document suggestion platform that provided automated recommendations for grammar, clarity, and professional tone, helping users produce polished business papers efficiently.
- Developed and maintained a Spring Boot backend with an SQL database, implementing RESTful APIs and optimizing query performance to reduce API response time by 35%.

PROJECTS

Recruiter Outreach Automation | Python, Flask, LangChain, OpenAI API, SQLite, Apollo.io API, Gmail API (OAuth2)

- Built a full-stack outreach automation tool that discovers company recruiters via the Apollo.io API and drafts personalized cold emails using OpenAI.
- Engineered a LangChain agent with a custom web-search tool to autonomously research target companies in real time, grounding each email in verifiable, company-specific facts instead of generic language.
- Designed a SQLite-backed review workflow with rate-limited sending to protect email deliverability and support safe, real-world outreach at scale.

Polymarket Trader (Live demo) | Next.js, FastAPI, PostgreSQL, Redis, AWS Lambda, Vercel, Render, Docker, Resend

- Built a full-stack automated trading platform that evaluates Polymarket prediction-market probabilities against user-defined rules, live across ~500 markets.
- Integrated the Alpaca API to automatically place corresponding stock trades based on triggered rules.
- Engineered an independently-scheduled evaluator on AWS Lambda (Docker/ECR, EventBridge) to execute rule and watchlist logic against a Postgres database.
- Implemented multithreading with Python's ThreadPoolExecutor to parallelize concurrent ticker price lookups, cutting refresh latency ~9x (0.89s → 0.10s).
- Implemented JWT authentication with bcrypt password hashing and Redis-backed caching for live market data.

TECHNICAL SKILLS AND CERTIFICATIONS

Languages: Java, Python, HTML, CSS, JavaScript, TypeScript, SQL, C, C++, MATLAB

Frameworks & Libraries: React, Node, TensorFlow, Keras, LangChain, Streamlit, Pandas, Hugging Face, XGBoost

Databases: MongoDB, PostgreSQL, Redis

DevOps & Infrastructure: Docker, AWS, Git, VSCode

Certifications: [AWS Certified Cloud Practitioner](#)